

Determining g-factors using Nuclear Magnetic Resonance

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Abstract

The existence of nuclear magnetic moments has allowed scientists to study the internal workings of atoms using electromagnetic methods. Nuclear magnetic resonance experiments take advantage of the atomic spin in order to determine the properties of the intrinsic magnetic fields that mediate interactions at the atomic level. This report explores the experimental methodology and properties of nuclear magnetic resonance, while finding the g-factors of a couple different samples.

1 Introduction

The nuclear magnetic moment is best described by quantum mechanics. In quantum physics, it is known that particles have an intrinsic angular momentum. Specifically, a proton can have an angular momentum $\mathbf{J}$. Scientists have observed that particles like the proton also have a magnetic moment $\boldsymbol{\mu}$ related to their angular momenta [1].

$$\boldsymbol{\mu} = \gamma \mathbf{J} = \frac{g\mu_N}{\hbar} \mathbf{J} \quad (1)$$

Where g is the dimensionless g -factor and μ_N is called the **nuclear magneton**. For an electron, one would replace the nuclear magneton with the **bohr magneton** μ_B in the above equation. In quantum physics, quantities like the angular momentum are quantized, and therefore the magnetic moment is quantized as well. The intrinsic angular momentum is also called the **spin** of the particle, and has an associated quantum number s :

$$\mathbf{J} = \hbar s$$

The value of s is dependent on the type of particle. Protons and electrons are Fermions, and therefore have spin $\frac{1}{2}$. In addition to a value, the spin of a particle has an orientation associated with a magnetic quantum number m_s . The magnetic quantum number determines the orientation and therefore the magnetic state of the particle, and can take on values $s, s-1, \dots, 0, -s+1, -s$. When a particle such as an electron or proton is placed into a magnetic field, it gains energy due to the magnetic moment. The energy of a magnetic moment in a static field is

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}_0$$

provided that $\boldsymbol{\mu}$ and $\mathbf{B}_0$ are oriented along the same axis. For simplicity and without loss of generality, we say that both are oriented along the z -axis. Due to the value of s for protons & electrons, $m_s = \pm \frac{1}{2}$. This demonstrates that once one of these particles is placed into an external magnetic field, the energy gain reveals two states, one with $m_s = \frac{1}{2}$ and the other with $m_s = -\frac{1}{2}$. These can roughly be translated to parallel and antiparallel to the magnetic field, although not exactly. The energies of the two states are

$$U_1 = -\frac{1}{2}\gamma\hbar B_0 \quad \text{and} \quad U_2 = \frac{1}{2}\gamma\hbar B_0 \quad (2)$$

Thus the energy difference between the magnetic states is

$$\Delta U = \gamma\hbar B_0 = g\mu_N B_0 \quad (3)$$

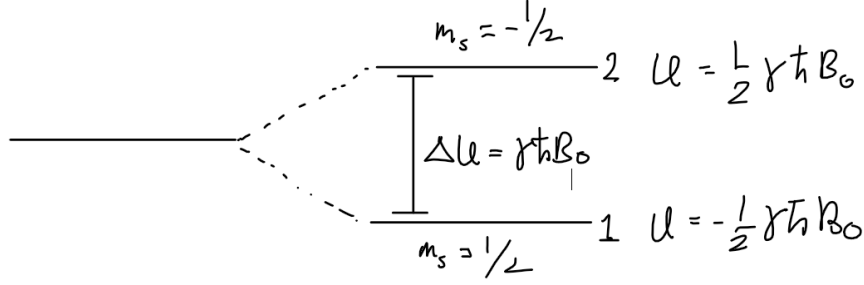


Figure 1: new energy states created by a static external magnetic field and quantized by m_s

Now, this is a good approximation, but most of the spins of the individual atoms are oriented slightly at an angle from the z -axis and therefore a static magnetic field in the z -direction will cause **precession** [2].

1.1 Magnetic Resonance

Magnetic precession is an interesting phenomenon, but it is difficult to measure and doesn't reveal a lot about the internal workings of the atoms. However, if we excite protons & electrons using radio pulses in the form of a transverse magnetic field, we can observe nonequilibrium behavior and different interactions inside of atoms. To apply these pulses, we apply an oscillating magnetic field perpendicular to the static magnetic field, which we will call $\mathbf{B}_1$. When this field is applied, the magnetic moments will begin to precess around it and the z -components of the magnetic moments will change.

If $\mathbf{B}_1$ oscillates at the same speed as the magnetic moments are precessing, a large maximum will be observed, as the z -components change for more time. This is called **resonance** and it essentially means that the driving frequency for the oscillation is the same as the oscillation frequency, giving an optimal output.

At the atomic level, photons must be the exact energy to excite protons to the higher energy state. To observe the maximum changes in the overall z -component of the total magnetic moment, we must excite the maximum amount of protons to the higher energy state. Thus the incoming photons (which are the radio pulses) must be the correct energy to excite the protons, which is given in equation (3).

To find the resonant frequency, one uses the Planck law, that energy is proportional to frequency, $E = \hbar\omega_0$. Thus we can find the resonant frequency using equation (3).

$$\omega_0 = \gamma B_0 = \frac{g\mu_N}{\hbar} B_0 \quad (4)$$

This will be the main equation used to find the g -factors.

The external static magnetic field $\mathbf{B}_0$ is meant to be constant across the sample, but it is in fact slightly inhomogeneous and has very slightly different values at different points in the sample. This means that individual proton magnetic moments are precessing at different speeds. In the experimental setup, the electrical signal is from the xy -component of the total magnetic moment, the component perpendicular to the static field. The inhomogeneity of the static field leads to the individual magnetic moments 'spreading out' as in Figure (2).

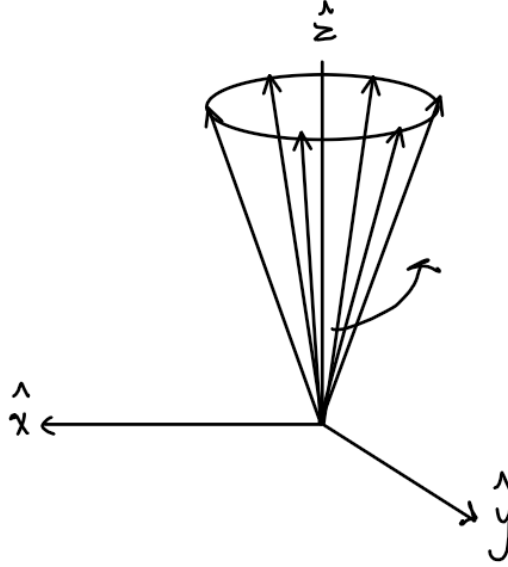


Figure 2: spreading out of precessing magnetic moments (arrows) due to the inhomogeneity of the static magnetic field, causing a decay of the xy signal

As can be seen the xy -component of the total magnetic moment (which is the sum of all the individual magnetic moment vectors) is effectively zero. The process and resulting signal due to this 'spreading out' is called the T_2^* decay, and will be measured in this experiment.

1.2 Absorption and Dispersion

When a photon with the correct frequency interacts with a particle there are two processes that can occur, and it depends on the initial state of the particle. If the particle is in the lower energy state (see Figure 1) then the photon will be absorbed and the particle will jump to the higher energy state. If the particle is in the higher energy state, it can fall to the lower energy state, emitting a photon of the resonant energy. The first process is called an **absorption**, and the second is called a **stimulated emission**.

The dominant process is dependent on the material, and as will be seen later in this report, the two materials we used will exhibit different electrical signals based on whether absorption or stimulated emission is taking place. The curve for a stimulated emission is called a dispersion curve [3].

2 Experimental Setup and Procedures

2.1 Magnetic Fields

The main experiment involved controlling and generating specific magnetic fields in order to obtain resonance. The static magnetic field, $\mathbf{B}_0$ was generated by a strong electromagnet, and the sample was placed inside.

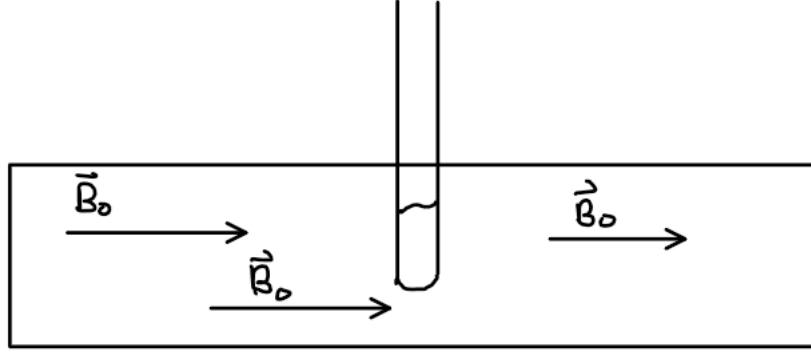


Figure 3: generation of static magnetic field $\mathbf{B}_0$

The above image (figure 3) shows the sample placed into the metal bar that contained the static magnetic field. This field could be controlled by a current supply, increasing the current would change the field. Figure 4 shows the measured relationship between current and the static field, as measured using a standard probe.

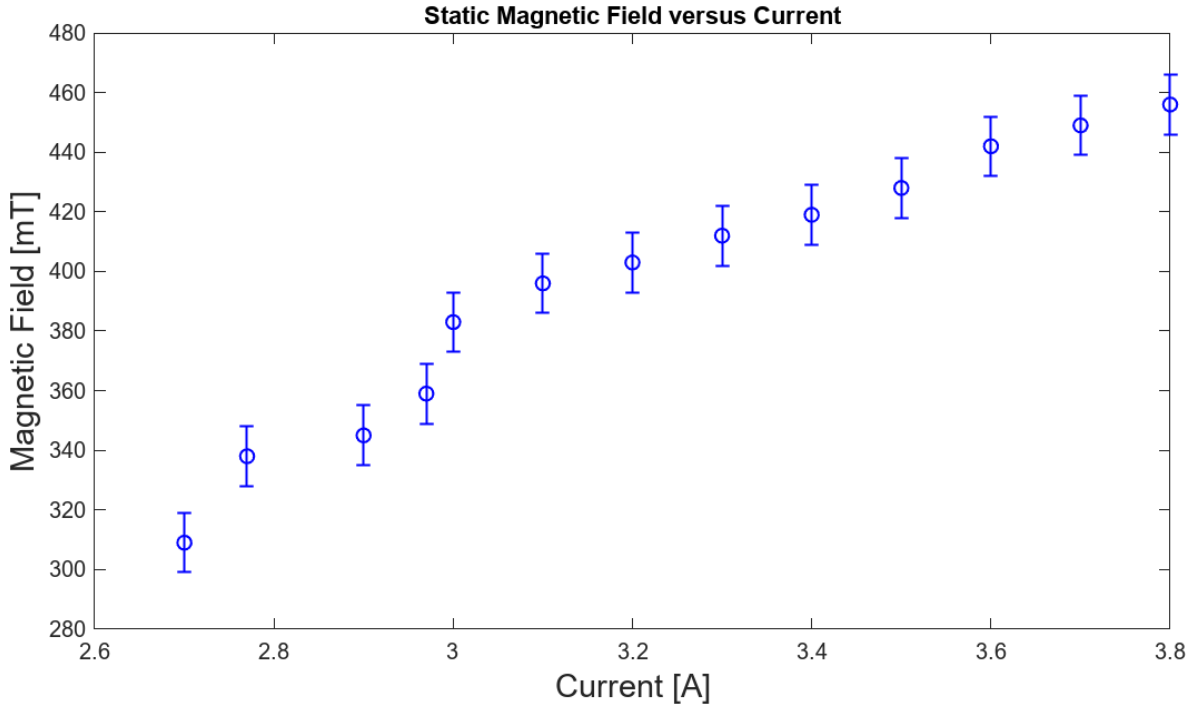


Figure 4: the static magnetic field $\mathbf{B}_0$ as generated by different currents

The above relationship will be useful when determining resonant frequencies. To observe electrical signals pertaining to resonance, we need to precess the magnetic moments around a magnetic field transverse to the sample and the bar, perpendicular to $\mathbf{B}_0$, so that we can measure the actual behavior. We called this field $\mathbf{B}_1$, and it was generated by an LC oscillator through a solenoid. If $\mathbf{B}_1$ oscillates back and forth at the resonant frequency, a resonance signal will be observed. Inside the metal bar there is a hole in which the sample is placed. Inside that hole is a coil of wire, or a solenoid. It is implied there is then a capacitor attached to create an LC oscillator.

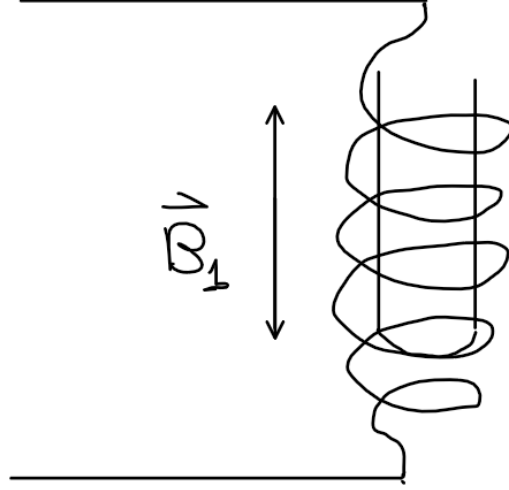


Figure 5: generation of the transverse oscillating field $\mathbf{B}_1$ in order to observe nuclear magnetic resonance

The frequency of $\mathbf{B}_1$ was adjusted throughout the course of the experiment to find the resonant frequency. As stated in the introduction, $\mathbf{B}_1$ will excite protons and electrons to their higher energy states, creating an observable signal. However, if $\mathbf{B}_1$ is the correct frequency for too long, more protons and electrons will keep getting excited and essentially overload the system. This means $\mathbf{B}_1$ can only be at the resonant frequency for short pulses before changing, and there are two ways to accomplish this [3].

The two factors affecting whether the sample is at resonance are the static field ($\mathbf{B}_0$), which controls the resonant frequency, and the oscillating field ($\mathbf{B}_1$) which is the radio frequency. To be at resonance the frequencies must equal each other and therefore changing either of the fields will affect the resonance. Since we wanted to use the radio frequency as a dependent variable, the setup uses the static field to prevent overloading.

By generating a third magnetic field $\mathbf{B}_S$ along the same axis as $\mathbf{B}_0$, we can change the static field and thus change the resonant frequency. By oscillating $\mathbf{B}_S$, the resonant frequency of the static field will only appear once every period of oscillation, and the signal will not overload. This process is called **sweeping** the resonant frequency.

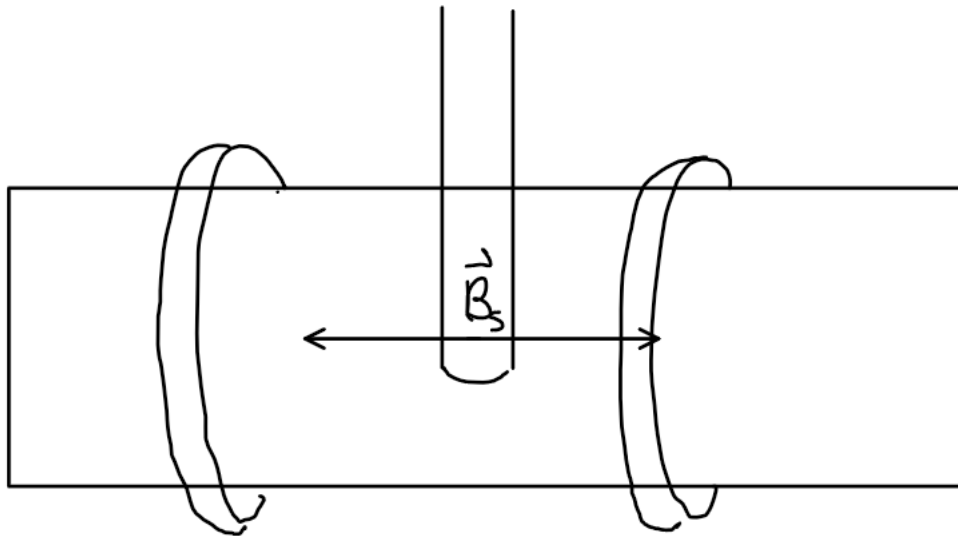


Figure 6: generation of $\mathbf{B}_S$ using Helmholtz coils

$\mathbf{B}_S$ was generated using Helmholtz Coils wrapped around the metal bar, as shown in Figure 6. The magnetic field of Helmholtz Coils is linearly related to the current supplied [3] so the supplied current was oscillated at a frequency of around 30 Hz. This is much slower than the resonant frequency (which is on the order of 15 MHz) and therefore did not affect the resonance signal in an observable way.

2.2 Sweeping

It is important to understand what exactly the sweeping process entails in order to interpret the output signals correctly. The apparatus oscillates the current through the Helmholtz coils, creating a magnetic field $\mathbf{B}_S$:

$$\mathbf{B}_S = B_S \sin(\omega t) \hat{z} \quad (5)$$

The resonant frequency can be calculated from the 'static' magnetic field (which is now oscillating due to $\mathbf{B}_S$).

$$\mathbf{B}_{\text{static}} = \mathbf{B}_0 + B_S \sin(\omega t) \hat{z} \quad (6)$$

ω is the 30 Hz frequency of the sweeping field discussed previously. To find the resonant frequency ω_0 , we use equation (4):

$$\omega_0 = \gamma B_0 + \gamma B_S \sin(\omega t) \quad (7)$$

Thus during the sweeping processes, the *resonant frequency* is changing with time. During this, we are changing the radio frequency to try and observe the resonance signals. Each time that the radio frequency and the resonant frequency are the same, we observe special behavior. On the oscilloscope, we observe the oscillation of $\mathbf{B}_S$ in one channel and the resonance detection in another. The oscillation of the sweeping field really is the oscillation of the resonant frequency. If, while the radio frequency ω_r is held constant, the resonant becomes equal to ω_r , the signal will be registered. Therefore we should be observing periodic resonance signals as ω_0 oscillates.

Suppose that the resonant frequency with just $\mathbf{B}_0$ is ω_0 , and the radio frequency is set to $\omega_r > \omega_0$. Now when we apply $\mathbf{B}_S$ to the static field, the resonant frequency oscillates through $\omega_0 \pm \omega_s$ where ω_s is determined by the amplitude of the $\mathbf{B}_S$ signal. If ω_r is within this range, then a resonance peak is observed when the resonant frequency reaches the right point. Since $\omega_r > \omega_0$, the peaks will only appear when the sweeping field is near the peak, on both sides of the peak. Therefore we will observe two resonance structures close to each other at each peak of the sweeping field.

This is critical to the experiment as the spacing between observed resonance peaks indicates where the input radio frequency is relative to the actual resonant frequency (the one without sweeping field). By changing the input radio frequency the spacing between the peaks can be changed, and therefore to measure the actual resonance frequency one must make sure the structures are space **evenly** so that the resonance is occurring when the sweeping field is zero.

2.3 Procedure

To observe nuclear magnetic resonance, two materials were observed, hydrogen 1H in a CuSO_4 aqueous solution, and fluorine (PTFE). The electromagnet (static field) was supplied a known amount of current, and the oscillating fields were started. Each sample was placed into the chamber and the radio frequency was tuned until resonance structures were observed. Once the peaks had been found, the radio frequency was finely tuned to evenly space the peaks. Once finished, the resonant frequency was recorded and uncertainty obtained by measuring the width of the resonance structure (more uncertainty explanation in section 4).

The supplied current (static field) was then incremented by small amounts to change the resonant frequency and the process was repeated. The end result was a relationship between the magnetic field and the resonant frequency, allowing us to calculate the g-factors. As per the theory (equation 4) this relationship should be linear, and the slope should contain the g-factor we need. Depending on if we use the nuclear magneton μ_N or the Bohr magneton μ_B we can calculate the g-factor for the protons or electrons involved.

3 Observations and Data

3.1 Hydrogen

The sample of 1H produced a dispersion curve on the oscilloscope.

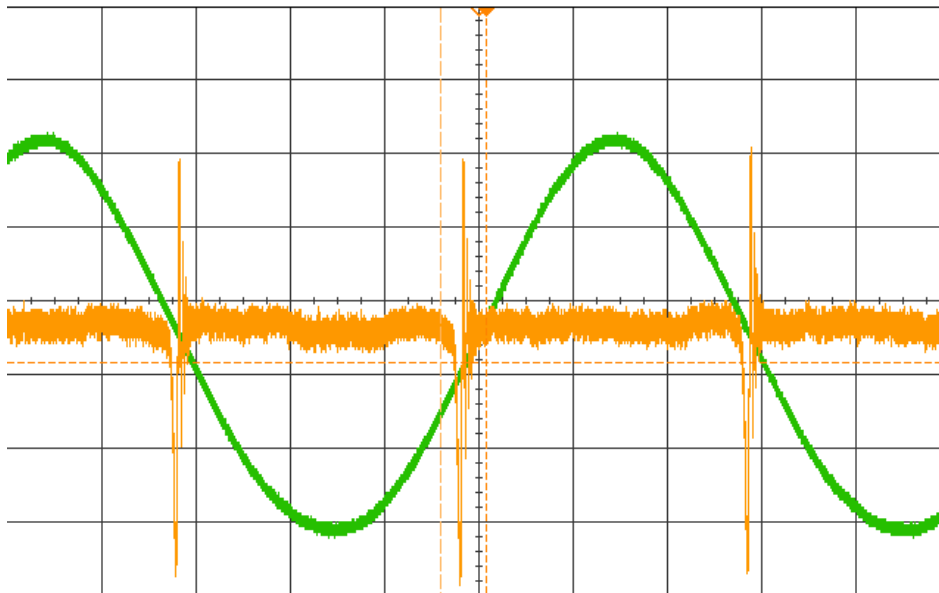


Figure 7: resonance structure of 1H excited by the radio frequency pulse. The 'noisy' signal represents the magnetic moment of the sample during resonance, and shows a clear dispersion curve. The sinusoidal curve represents the sweeping magnetic field.

In Figure (7) one can observe the resonance peaks evenly spaced so that the radio frequency is the natural resonant frequency of the sample. 1H is really just individual protons and electrons, so the observation of this sample gives a good idea of the g factor for a single protons as well. As stated in section 2.3 this state was recorded for multiple different currents (meaning multiple magnetic fields).

Current [A] ± 0.01	Magnetic Field [mT] ± 1	Resonant Frequency [MHz] ± 0.001
2.7	309	15.733
2.77	338	16.046
2.97	356	16.876
3.2	403	17.883
3.4	419	18.653

Table 1: resonant frequencies for different magnetic fields in order to find the g-factor for protons

for a closer look at the shape of the dispersion curve, see figure (8).

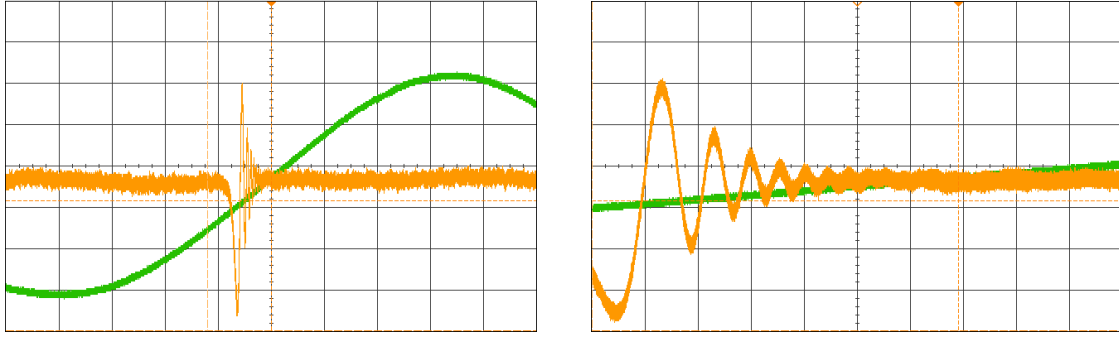


Figure 8: zoomed in views on the resonant structure for protons in CuSO_4 , showing distinct oscillatory behavior as well as exponential decay. Again, the larger sinusoid represents the sweeping magnetic field

3.2 Fluorine

Similar to the previous section, we observed the resonance structures for fluorine atoms in the sample of PTFE.

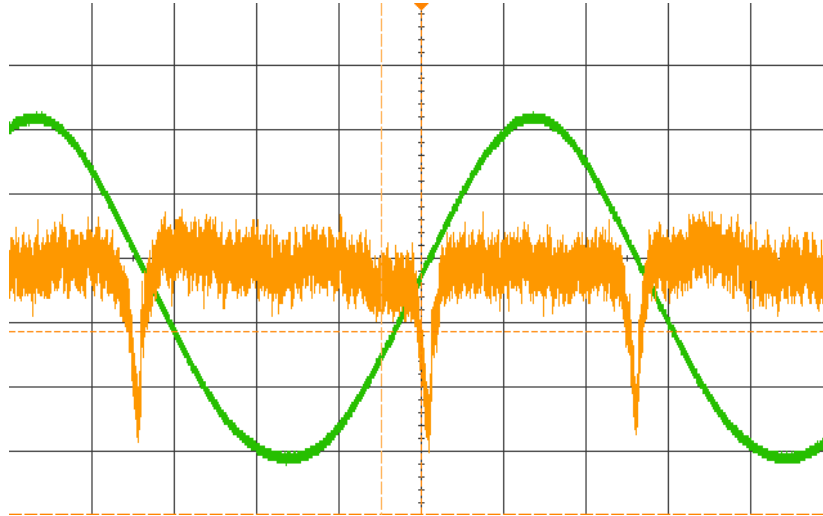


Figure 9: scope trace of the resonant structures for fluorine, showing clear absorption shapes.

In Figure (9) one can see the absorption structure of the Fluorine resonance. The dip shows that lots of individual atoms are transitioning to the higher energy levels.

Current [A] ± 0.01	Magnetic Field [mT] ± 1	Resonant Frequency [MHz] ± 0.001
3.0	383	16.045
3.1	396	16.495
3.2	403	16.844
3.3	412	17.201
3.4	419	17.549
3.5	428	17.921
3.6	442	18.136
3.7	449	18.449
3.8	456	18.747

Table 2: measured resonant frequencies at different values of the static magnetic field

4 Analysis & Discussions

The first item to discuss is the uncertainty in the resonant frequency. In tables it was reported to be ± 0.001 MHz, but this is just the uncertainty that the equipment reported. Theoretically the resonant frequency is at the peak of the signal, but since it is a peak, the width can give a good estimate for the uncertainty. The absorption peaks are more well defined so we measured the widths of multiple absorption peaks and averaged to find that on average the peaks had a width of 1.3 ms. This, however, is not in the right units, as the x axis on the scope represents time. To convert to units of frequency we needed to find a transformation to the frequency.

This was matter of finding the resonant frequency as a function of time. Since the peaks were tuned to be centered where the sine wave was zero, it is reasonable to find the sine function representing the frequency and apply it centered around $t = 0$. The frequency of the sweeping field was measured to be around 33 ± 1 Hz by the oscilloscope, and by measuring the frequencies at the peaks of the sine wave the amplitude was found to be 0.115 ± 0.001 MHz. Thus the equation was found:

$$\omega_0(t) = 115000 \sin(66\pi t) \quad (8)$$

Using this the uncertainty of a peak centered at $t = 0$ with a width of 1.3 ms was found to be 15.5 kHz.

4.1 g-factors & Systematic Error

Equation (4) shows how we may determine the g-factors for specific materials. A linear relationship between the resonant frequency and the static magnetic field was fit from the data and used to find the required value. Equation (4) is in terms of angular frequencies, but it can be transformed to become:

$$f_0 = \frac{g\mu}{h} B_0 \quad (9)$$

Where the reduced Planck constant has also been transformed to the regular Planck constant. The detector was not measuring angular frequencies so equation (9) is useful.

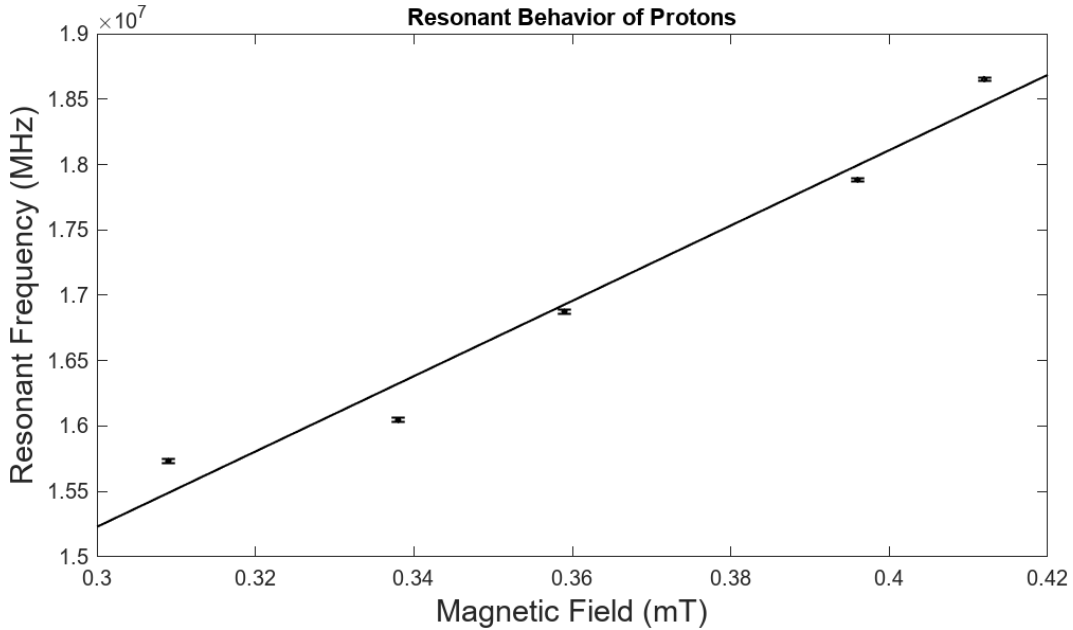


Figure 10: resonant behavior of protons including a linearly fitted relationship used to find the g factor

With 68% confidence intervals, this linear fit provided a g-factor of 3.8 ± 0.5 for the protons. A similar plot was created for the fluorine sample.

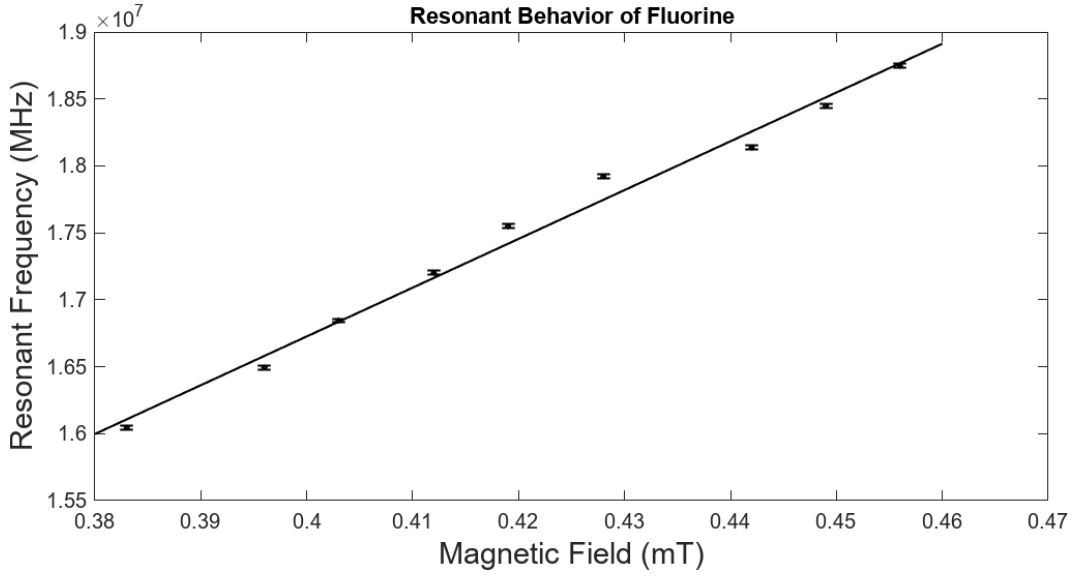


Figure 11: resonant behavior of fluorine including a linearly fitted relationship used to find the g factor

For fluorine, the linear fit provided a g factor of 4.8 ± 0.5 for the nucleus. In addition to linear fits, standard deviation of the mean (SDOM) analyses were performed for each data set.

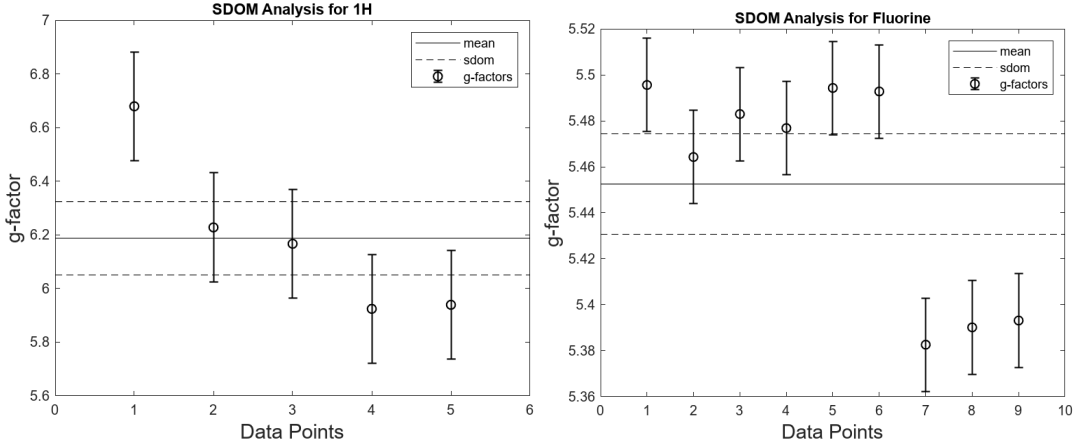


Figure 12: SDOM Analysis of g-factors for 1H and fluorine

These SDOM analyses were quite insightful as they produced different values than those calculated previously. This can be attributed to the significant y-intercept present in the linear fits. The average for 1H was 6.2 ± 0.1 and the average for fluorine was 5.45 ± 0.02 .

The accepted g-factor for a proton is $g_p = 5.586$ [4]. For 1H, we measured 3.8 for the linear fit and 6.2 for the SDOM analysis. Clearly the SDOM analysis was closer, with a percent difference of about 11%. For the fluorine the case was the same, the SDOM analysis was very close to the accepted value of the proton, with a 2.4% difference between the values.

These values are decently close, but if one applies a correction to f and to B in these situations, the SDOM g-factors can be brought closer to the accepted values. By subtracting 1 MHz off the frequencies and subtracting 10 mT from the magnetic fields, one can obtain a value of 5.98 from the 1H data and 5.27 from the fluorine data, both very close to the accepted value. This implies the presence of a systematic error in the process, meaning an external or environmental factor that affects all of the data in the same way. This could be an environmental magnetic field (that of the Earth or other equipment) or even the interactions between the atoms and nuclei themselves. Therefore the experimentally determined f and B are slightly off due to environmental factors.

4.2 Free Induction Decay

As discussed in section 1.1, the inhomogeneity of the static magnetic field produces a decay signal with a time constant denoted T_2^* . When the sample is in resonance, we get a large signal as the total magnetic moment begins to gain an xy-component which is measured on the scope. However this signal decays very fast due to the spreading out discussed in 1.1, and that time is measurable. Figure 8 shows a picture (on the right side) of this T_2^* signal, as the peaks of the resonant structure decay exponentially. Extracting this data and fitting to an exponential yields the time constant we are looking for.

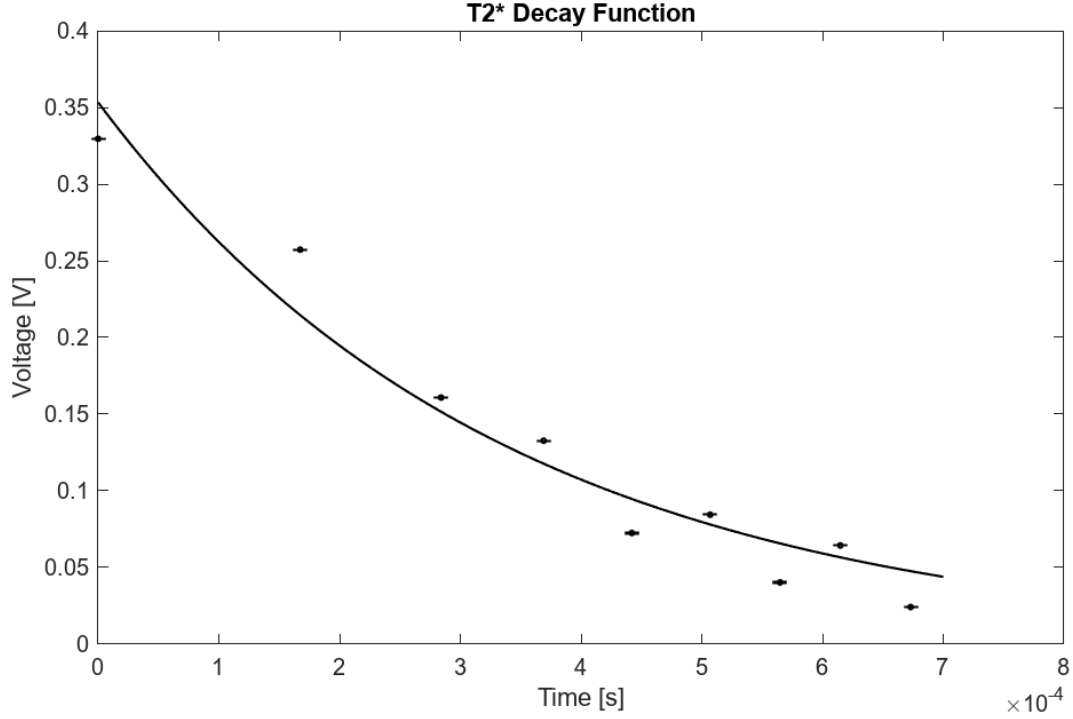


Figure 13: T_2^* function as an exponential decay, to find the time constant

The time constant as determined from this fit is $3.36 \pm 0.098 \times 10^{-4}$ s. The oscillations in the decay are due to the sweeping magnetic field which changes the resonant frequency and thus the output. When examining the molecular structures of the materials, CuSO_4 has a much smaller and more symmetric structure although it has a copper ion attached on one end. The PTFE structure resembles a long chain and is therefore very polar and has interactions with magnetic fields at the molecular level.

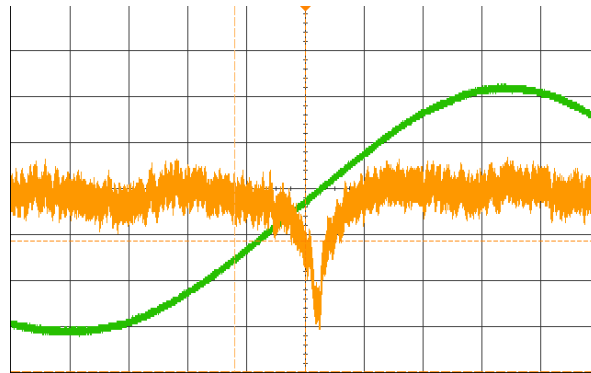


Figure 14: close up view of the resonant structure for PTFE, showing an absorption shape

Figure 14 shows a nicer view of the absorption curve for PTFE. This occurs because the dominant

process in PTFE is absorption, while the dominant process in the CuSO_4 is stimulated emission.

5 Conclusion

Interestingly, the g-factors for PTFE and CuSO_4 are very similar, even though their resonant structures look very different on the scope. As discussed in 1.2, this is simply due to the process being different, but with the same energy levels. The energy levels determine the value and the process (absorption or emission) determines the shape of the curve.

Nuclear magnetic resonance is a phenomenon that is difficult to capture, but rewarding to explore. The measurement apparatus consists of multiple oscillating magnetic fields and circuitry that probe the magnetic properties of a sample. An incoming radio pulse can only excite particles to a higher energy level if the incoming photon has exactly that energy, meaning each sample has a resonance frequency based on the splitting of the states. In other words, the resonant frequency is dependent on the external magnetic field, and therefore can be measured.

The g-factors are proportionality constants that relate the resonant frequency to the magnetic field, and using electromagnetic equipment we were able to measure the g-factors for two samples. The physics of nuclear magnetic resonance is essentially quantum mechanics combined with a classical interpretation of rotational mechanics. The classical interpretation was explored in a different experiment [2] but the results from that report are useful in determining the behavior of a single magnetic moment. In a sample however, statistical mechanics and quantum physics take over and change the system's behavior by defining quantized states and the total magnetic moment.

Nuclear magnetic resonance provides researchers with the ability to probe the interactions of fundamental atomic particles and understand the behavior of matter on an atomic level. This ability has applications in the fields of physics, chemistry, imaging, medicine, and many more.

References

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